

Certificate Validator

Complete Project Structure & Technical Architecture Specification Document

1. Executive Overview & System Architecture

The **Certificate Validator** system is a state-of-the-art, end-to-end cryptographic certificate issuance, management, verification, and decentralized privacy ecosystem. Built using Node.js/Express, SQLite/Better-SQLite3, and React (Vite, Framer Motion, HTML5 Canvas), it leverages RSA-2048 digital signatures, SHA-256 cryptographic hashing, Replay Attack Session Protection, Merkle-tree Blockchain Anchoring, Zero-Knowledge Selective Disclosure Predicates, and an offline Progressive Web Application (PWA) architecture.

Core Technology Stack

Layer	Technologies & Libraries Used	Primary Architectural Role
Frontend Core	React 18, Vite 8, React Router v7, Lucide Icons	Responsive SPA client with dynamic route switching and PWA caching
UI & Styling	Vanilla CSS, Glassmorphism Design Tokens, Framer Motion	Ultra-premium aesthetic design system with micro-animations & layout fades
Backend Core	Node.js, Express.js, Better-SQLite3, CORS, Multer	RESTful API server, SQLite database access, file uploads, & auth middleware
Cryptography	Node.js Crypto (RSA-2048, SHA-256), Web Crypto API	Asymmetric signatures, tamper checking, and browser-side offline verification
Templating	HTML5 Canvas, Lucide-React, ReportLab Python PDF Engine	Resolution-independent certificate rendering, PNG export, and PDF generation
AI Subsystem	Gemini API (Google AI), Custom Floating AI Assistant	Interactive platform assistant, verification guidance, & natural language QA
Excel / Bulk	XLSX (SheetJS), Custom Alias Mapper & Category Normalizer	Mandatory column assertion, row validation, and multi-cert batch generation

2. High-Level Project Directory Hierarchy

```
d:\Certificate_Validator\ ■■■ backend\ # Node.js Express REST Backend & Database Layer
■ ■■■ config\ # SQLite Database Connection & Table Schema Setup
■ ■■■ controllers\ # API Controllers (Auth, Certificate, Verifier, Analytics, etc.)
■ ■■■ middleware\ # JWT Auth Guard & User Role Verification Middlewares
■ ■■■ models\ # Data Access Models (User, Certificate, Verification, Wallet)
■ ■■■ routes\ # Express API Route Registrations
■ ■■■ services\ # Core Business Logic (Key Gen, QR Generator, PDF Export)
■ ■■■ tests\ # Automated Backend Test Suite (Crypto, Replay, Revocation)
■ ■■■ uploads\ # Server-side Generated QR Code Images & Cert Artifacts
■ ■■■ utils\ # Cryptographic Engine, Audit Logger, Blockchain Anchor
■ ■■■ database.sqlite # SQLite Relational Database Storage
■ ■■■ package.json # Backend Dependencies & Scripts Manifest
■ ■■■ server.js # Express Application Server Entry Point & Middlewares ■
■■■ frontend\ # React Vite Single Page Application & PWA Client
■ ■■■ public\ # PWA Manifest, Web App Icons, SVG Graphics
■ ■■■ src\ # React Application Source Code
■ ■ ■■■ api\ # Axios API Clients for Backend Communication
■ ■ ■■■ components\ # UI Components, Layouts, Visual Decor, Templates
■ ■ ■ ■■■ AIChat\ # Floating AI Chatbot Assistant & Modal Components
■ ■ ■ ■■■ decorations\ # Background Glassmorphism & Abstract Math Shapes
■ ■ ■ ■■■ motion\ # Framer Motion Wrappers (CountUp, Skeleton, Scroll)
■ ■ ■ ■■■ templates\ # 9 High-Res Certificate Templates & TemplateSelector
■ ■ ■ ■■■ wallet\ # Digital Skill Wallet Cards, Timeline, & Detail Panes
■ ■ ■■■ hooks\ # Custom React Hooks (Header Height, AI Provider, Scroll)
■ ■ ■■■ pages\ # 19 Page Components (Dashboard, Verifier, Wallet, Analytics)
■ ■ ■■■ styles\ # Wallet Specific CSS & Theme Extensions
■ ■ ■■■ utils\ # Excel Parser, PDF Generator, Offline Crypto, Key Cache
■ ■ ■■■ App.jsx # Application Navigation Router & Layout Provider
■ ■ ■■■ index.css # Global CSS Design Tokens, Glass Effects, Animations
■ ■ ■■■ main.jsx # React DOM Mount & Service Worker Registration
■ ■■■ index.html # Root HTML Document & Font Imports
■ ■■■ package.json # Frontend Dependencies & Build Configuration
■ ■■■ vite.config.js # Vite Bundler & PWA Plugin Configuration ■
■■■ docs\ # Platform Documentation Assets & Template Previews
■ ■■■ screenshots\ # UI Screenshots & Architectural Diagrams
■ ■■■ templates\ # Reference High-Res Template Image Assets (01 to 09) ■
■■■ package.json # Root Monorepo Management Manifest
■■■ README.md # Complete Project Documentation & Installation Guide
```

3. Detailed Folder Breakdown & Component Architecture

Folder Path	Module Name	Functional Description & Architectural Role
backend/config/	Database Configuration Layer	Contains SQLite database initialization (<code>db.js</code>). Automates table creation for users, universities, certificates, revoked certificates, blockchain anchors, selective disclosure proofs, and verification events.
backend/controllers/	API Request Controllers	Houses request handlers for auth (<code>authController.js</code>), single & bulk certificate issuance (<code>certificateController.js</code>), online verification (<code>verificationController.js</code>), selective disclosure (<code>disclosureController.js</code>), skill passports (<code>passportController.js</code>), audit logs (<code>auditController.js</code>), analytics (<code>analyticsController.js</code>), and AI chat assistant (<code>chatController.js</code>).
backend/middleware/	Security & Authorization Guards	Contains JWT authentication middleware (<code>authMiddleware.js</code>) that verifies authorization tokens, decodes user payload, and enforces role-based access control (UNIVERSITY, STUDENT, VERIFIER, ADMIN).
backend/models/	Data Access Layer	Direct SQLite database interface models (<code>userModel.js</code> , <code>certificateModel.js</code> , <code>verificationModel.js</code> , <code>walletModel.js</code>) providing parameterized SQL queries for atomic transactions and data integrity.
backend/routes/	Express API Routes	Maps REST HTTP endpoints (<code>authRoutes.js</code> , <code>certificateRoutes.js</code> , <code>verificationRoutes.js</code> , <code>disclosureRoutes.js</code> , <code>passportRoutes.js</code> , <code>auditRoutes.js</code> , <code>analyticsRoutes.js</code> , <code>chatRoutes.js</code>) to controller handlers.
backend/services/	Core Business Services	Encapsulates cryptographic key generation (<code>keyService.js</code>), high-density QR code image synthesis (<code>qrService.js</code>), server-side PDF compilation (<code>pdfService.js</code>), and wallet data processing (<code>walletService.js</code>).
backend/tests/	Automated Backend Test Suite	Node.js test suite (<code>crypto.test.js</code>) covering RSA-2048 keypair generation, SHA-256 hash checks, signature verification, replay protection nonces, revocation anchoring, and selective disclosure predicates.
backend/utils/	Cryptographic & Audit Utilities	Contains core crypto utilities (<code>crypto.js</code>) for RSA/SHA256, blockchain anchoring simulation (<code>blockchain.js</code>), and immutable audit trail logging (<code>auditLogger.js</code>).
frontend/src/api/	Axios Network API Clients	Modular API modules (<code>client.js</code> , <code>analytics.js</code> , <code>audit.js</code> , <code>blockchain.js</code> , <code>chat.js</code> , <code>disclosure.js</code> , <code>passport.js</code> , <code>templateApi.js</code> , <code>wallet.js</code>) providing seamless backend REST communication with automatic bearer token injection.
frontend/src/components/templates/	High-Resolution Certificate Templates	9 individual visual certificate templates (<code>GraduationTemplate</code> , <code>CourseCompletionTemplate</code> , <code>InternshipTemplate</code> , <code>ProjectTemplate</code> , <code>MeritTemplate</code> , <code>DistinctionTemplate</code> , <code>AcademicExcellenceTemplate</code> , <code>BonafideTemplate</code> , <code>ParticipationTemplate</code>) and dynamic category router <code>TemplateSelector.jsx</code> .

frontend/src/components/wallet/	Digital Skill Wallet Components	Components for student wallet interface (`WalletCertCard`, `WalletDetailPane`, `WalletIndexList`, `CertDetailModal`, `AchievementTimeline`, `LearningGoalTracker`, `PortfolioLinksCard`).
frontend/src/components/AIChat/	Floating AI Assistant Components	Interactive AI Chatbot assistant widget (`FloatingAIButton`, `ChatWindow`, `ChatBubble`, `SuggestedQuestions`, `TypingIndicator`) connecting to Gemini API.
frontend/src/pages/	Application Page Views	19 page components handling University Dashboard (`UniversityDashboard.jsx`), Certificate Verifier (`Verifier.jsx`), Student Wallet (`WalletDashboard.jsx`), Skill Passport (`DigitalSkillPassport.jsx`), Analytics, Audit Logs, and Auth.
frontend/src/utlis/	Client Utilities & Off-line Engines	Client-side Excel parsing (`excelParser.js`), category normalization (`certificateCategory.js`), browser PDF generator (`certificatePdf.js`), QR decoder (`qrDecoder.js`), offline RSA crypto (`offlineCrypto.js`), and revocation caching (`revocationCache.js`).

4. Repository Summary Metrics

Metric Description	Value / Count
Total Backend Core Source Files	28 Files
Total Frontend Core Components & Pages	54 Component & Page Files
Certificate Visual Templates Supported	9 Category Templates (Vertical & Horizontal)
Automated Backend Unit Tests	13 Test Suites (100% Pass Rate)
Cryptographic Algorithms Enforced	RSA-2048 PKCS#1 v1.5, SHA-256, Merkle Tree
Database Schema Tables	8 Relational Tables with Foreign Key Constraints